

HYEON LEE

hyeonlee@unc.edu | <https://hyeonlee-stats.github.io>

EDUCATION

University of North Carolina at Chapel Hill Aug. 2022 - Present (Expected May 2027)
Ph.D. Candidate in Statistics and Operations Research, advised by Professor J. S. Marron

Seoul National University Mar. 2020 - Feb. 2022
M.S. in Statistics, advised by Professor Byeong U. Park

Korea Advanced Institute of Science and Technology Mar. 2015 - Feb. 2020
B.S. in Mathematical Science, minor in Industrial and Systems Engineering

HONORS AND AWARDS

Outstanding Achievement Award Dec. 2023
Department of Statistics and Operations Research, UNC Chapel Hill

Presidential Science Scholarship 2015 - 2019
Korea Student Aid Foundation, Republic of Korea

RESEARCH INTERESTS

Principal Subsimplex Analysis

Developing statistical methodologies for dimension reduction and identification of mode of variation for compositional data via nested sequence of subsimplices.

Functional and Shape Analysis of Biomechanical Curves

Analyzing regression models for gait curves with repeated measurements using Elastic Shape Analysis.

Functional Analysis using Wasserstein-Fisher-Rao Distance

Developing a workflow for analyzing functional data in the tangent space of the space of measures with the Wasserstein-Fisher-Rao distance.

Data Integration for Numerical and Categorical Variables

Studying an extension of Data Integration Via Subspace Analysis (DIVAS) for categorical variables.

PUBLICATIONS & PREPRINTS

Peer-Reviewed Publications

Lee, H., Hingee, K. L., Scealy, J. L., Wood, A. T., Grunsky, E., and Marron, J. S. (2026). “Principal subsimplex analysis”. *Journal of computational and graphical statistics*. accepted, pp. 1–18.

Buck, A. N., Borgert, J., **Lee, H.**, Arbeeva, L., Golightly, Y. M., Loeser, R. F., Messier, S. P., Pietrosimone, B., Nelson, A. E., and Marron, J. (2025). “Vertical ground reaction force variability is associated with clinical features in individuals with knee oa and overweight/obesity: a novel machine learning analysis of the IDEA trial”. *Osteoarthritis and cartilage* 33, S160–S161.

Submitted Papers & Preprints

Lee, H., Marron, J. S., Grunsky, E., and Scealy, J. “Dimension reduction of compositional data into mixtures of latent source compositions”. In preparation.

Lee, H. and Marron, J. S. “Analysis of nonnegative curves using Wasserstein-Fisher-Rao distance”. In preparation.

Lee, H., Armitano-Lago, C., Kelly, D. K., Arbeeve, L., Borgert, J., Buck, A. N., Messier, S. P., Nelson, A. E., and Marron, J. S. “Disentangling the effects of diet, exercise, walking speed, and weight loss on vertical ground reaction forces in knee osteoarthritis: findings from the IDEA trial”. Submitted at Journal of Applied Biomechanics.

Buck, A. N., Borgert, J., **Lee, H.**, Lisee, C. M., Li, X., Favoreto, N., Arbeeve, L., Callahan, L. F., Loeser, R. F., Thoma, L., Marron, J. S., Nelson, A. E., and Pietrosimone, B. “Machine learning identifies biomechanical and biological phenotypes after acl reconstruction that are linked to worse cartilage composition”. In preparation.

SOFTWARE

psacomp (R package), available at GitHub. (<https://github.com/hyeon-lee-stat/psacomp>)

ACADEMIC ACTIVITIES, WORKSHOPS & PRESENTATIONS

Invited & Contributed Presentations

- “Principal Subsimplex Analysis”, Joint Statistical Meetings (Contributed Paper), Nashville, TN, 2025
- “Principal Subsimplex Analysis”, Object Oriented Data Analysis and its Applications (Poster Presentation), UNC Chapel Hill, Chapel Hill, NC, 2025

Workshops & Summer Schools

- “Introductory School: Geometric Statistics”, Hausdorff Research Institute for Mathematics, Bonn, Germany, 2026 (Travel Support Fellowship Received)

TEACHING

Teaching at UNC Chapel Hill

- Instructor for STOR 155: Introduction to Data Models and Inference Fall 2025
- Teaching assistant for STOR 155: Introduction to Data Models and Inference Fall 2023
- Teaching assistant for STOR 535: Probability for Data Science Spring 2023
- Teaching assistant for STOR 435: Introduction to Probability Fall 2022

Teaching at Seoul National University

- Teaching assistant for 033.020 Statistics Lab. Fall 2020, Spring 2021, Fall 2021